Inflammatory Bowel Disease (IBD)

Ulcerative colitis and Crohn's disease — all key points for the exam + drug summary table

1. Overview: Ulcerative Colitis vs Crohn's Disease

IBD = **chronic, relapsing** immune-mediated inflammation of the GI tract, with periods of **flare** and **remission**. Two main types: **ulcerative colitis (UC)** and **Crohn's disease (CD)**.

Feature	Ulcerative colitis (UC)	Crohn's disease (CD)
Location	**Colon and rectum only** • Always starts in the **rectum** and extends upward	**Anywhere from mouth to anus** • Most often the **terminal ileum** (ileocecal)
Pattern	**Continuous** (no skip areas)	**Skip lesions** (patchy) • **Cobblestone** appearance
Depth	**Mucosa** (superficial)	**Transmural** (full thickness)
Main symptoms	**Bloody diarrhea**, urgency, tenesmus	**Abdominal pain** (right lower quadrant), diarrhea (less bloody), **weight loss**, fatigue
Complications	**Toxic megacolon**, severe bleeding, colon cancer	**Fistulas**, **strictures** (obstruction), **abscesses**, **perianal disease**, **malabsorption**, colon cancer
Granulomas	No	**Yes** (non-caseating)
Smoking	**Protective** (smoking ↓ risk; but never recommended)	**Worsens disease** → **stop smoking** (critical)
Antibodies	**p-ANCA** positive	**ASCA** (anti-Saccharomyces) positive
Surgery	**Colectomy is curative**	Surgery is **not curative** (disease recurs)
Malabsorption	Uncommon	**Common** (ileal disease/resection): **B12**, iron, **fat-soluble vitamins**, bile acids, zinc, folate
Liver	**Primary sclerosing cholangitis (PSC)** more common	Gallstones

2. Extraintestinal Manifestations

System	Manifestations
Joints	**Arthritis** (most common), **ankylosing spondylitis**, sacroiliitis
Skin	**Erythema nodosum** (follows disease activity), **pyoderma gangrenosum**, aphthous mouth ulcers
Eyes	**Uveitis**, **episcleritis**
Liver	**Primary sclerosing cholangitis** (UC) → ↑ colon cancer and cholangiocarcinoma risk
Kidney	**Calcium oxalate kidney stones** (CD with ileal disease/fat malabsorption), uric acid stones
Blood	**Anemia** (iron deficiency, chronic disease, B12), **VTE** (2–3 × risk; higher during flares and hospitalization)
Bone	**Osteoporosis** (steroids, inflammation, vitamin D/calcium malabsorption)
Growth	**Growth failure** and delayed puberty in children (CD)

3. Disease Extent and Severity

UC extent (Montreal classification)

Extent	Area	Treatment route
Ulcerative proctitis	**Rectum only**	**Suppository** (reaches ~15–20 cm)
Left-sided (distal) colitis	Up to the **splenic flexure**	**Enema** (reaches the splenic flexure) ± oral
Extensive colitis / pancolitis	Beyond the splenic flexure	**Oral** (± rectal)

UC severity (Truelove and Witts / ACG)

Severity	Features
Mild	**< 4 stools/day**, little blood, no systemic signs, normal ESR/CRP
Moderate	**4–6 stools/day**, moderate blood, minimal systemic signs
Severe	**≥ 6 bloody stools/day** + **systemic toxicity**: **HR > 90**, **temp > 37.8 °C**, **Hgb < 10.5**, **ESR > 30** (or ↑ CRP)
Fulminant	> 10 stools/day, continuous bleeding, transfusion need, **toxic megacolon**

CD severity

- **Crohn's Disease Activity Index (CDAI): < 150** remission • **150–220** mild–moderate • **220–450** moderate–severe • **> 450** severe–fulminant.
- **Behavior:** inflammatory, **stricturing**, or **penetrating (fistulizing)** ± **perianal** disease.
- **High-risk features** (→ early biologics): young age at diagnosis, **perianal** disease, extensive disease, deep ulcers, prior surgery, stricturing/penetrating behavior.

Monitoring markers

Fecal calprotectin (intestinal inflammation), **CRP**, ESR, CBC, albumin, endoscopy (mucosal healing). Always test for **C. difficile** (and other infections) during a flare.

4. Treatment Goals and Principles

- **Induce remission** (treat the flare) → **maintain remission** (prevent relapse).
- **Steroid-free remission** and **mucosal (endoscopic) healing** ("treat-to-target").
- Prevent complications, hospitalization, and surgery; improve quality of life.
- **Corticosteroids are for INDUCTION only — never for maintenance.**
- **Thiopurines and methotrexate are slow → MAINTENANCE** (steroid-sparing), not induction alone.
- A **biologic or small molecule that induces remission** is usually **continued for maintenance**.
- **Aminosalicylates (5-ASA)** work for **UC**, but **not** for CD.

5. Aminosalicylates (5-ASA)

- **Mechanism: topical anti-inflammatory** effect on the gut mucosa (↓ prostaglandins, leukotrienes, cytokines). Must reach the inflamed area → formulations differ in **where they release**.
- **Use: first-line for mild–moderate UC** (**induction and maintenance**). **Not recommended for Crohn's disease**.
- **Rectal 5-ASA** is **more effective than rectal steroids** for distal UC; **oral + rectal combination** is better than oral alone.
- May **reduce colorectal cancer** risk in UC.
- **Safe in pregnancy** and breastfeeding.

Drug	Release site / mechanism	Dose
Mesalamine delayed-release (Asacol HD, Delzicol)	pH-dependent → **terminal ileum and colon**	Induction **2.4–4.8 g/day** • Maintenance **≥ 2 g/day**
Mesalamine MMX (Lialda)	Multimatrix → **whole colon • once daily**	**2.4–4.8 g once daily**
Mesalamine extended-release (Apriso)	Granules, colon	1.5 g once daily (maintenance)
Mesalamine controlled-release (Pentasa)	Ethylcellulose microgranules → **small intestine and colon**	2–4 g/day in divided doses
Mesalamine suppository (Canasa)	**Rectum**	**1 g at bedtime — first-line for proctitis**
Mesalamine enema (Rowasa)	**Up to the splenic flexure**	**4 g at bedtime** (retain ≥ 8 hours) — left-sided colitis
Sulfasalazine (Azulfidine)	**Prodrug**: 5-ASA + **sulfapyridine**, split by **colonic bacteria** → colon	Induction **4–6 g/day** divided • Maintenance **2–4 g/day** • Start low, titrate
Balsalazide (Colazal)	Prodrug → colon (bacterial azoreduction)	**2.25 g TID** (6.75 g/day)
Olsalazine (Dipentum)	Two 5-ASA molecules → colon	500 mg BID • **Secretory diarrhea**

Adverse effects

Drug	ADRs
Sulfasalazine (mostly from **sulfapyridine**)	**Nausea**, vomiting, **headache**, anorexia (dose-related) • **Rash**, **sulfa allergy** • **Hemolytic anemia** (**G6PD deficiency**) • **Agranulocytosis**, bone marrow suppression • **Hepatotoxicity** • **Reversible male infertility (oligospermia)** • **Orange-yellow urine and skin**, stains contact lenses • **Folate deficiency** → give **folic acid 1 mg daily** (2 mg in pregnancy) • Monitor CBC and LFTs
Mesalamine (better tolerated)	**Headache**, nausea, abdominal pain, diarrhea • **Interstitial nephritis** → **monitor SCr** • **Pancreatitis**, hepatitis, pericarditis/myocarditis (rare) • **Paradoxical worsening of colitis** (acute intolerance syndrome) • Avoid in **salicylate allergy**

Drug interactions

- **Sulfasalazine:** ↓ **folate** absorption, ↓ **digoxin** absorption • ↑ **myelosuppression with thiopurines** (TPMT inhibition).
- **Mesalamine/sulfasalazine + azathioprine/6-MP:** ↑ risk of **myelosuppression**.
- **pH-dependent mesalamine** (Asacol HD, Delzicol, Lialda): **antacids, PPIs, H2RAs** may cause early release.

6. Corticosteroids (induction only)

Drug	Use	Dose
Budesonide MMX (Uceris tablet)	**Mild–moderate UC** induction (when 5-ASA fails)	**9 mg daily × 8 weeks**
Budesonide EC (Entocort EC)	**Mild–moderate ileocecal (ileal/right colon) CD** induction (**first-line**)	**9 mg daily × 8 weeks** • May use 6 mg daily for up to 3 months
Budesonide rectal foam (Uceris foam)	Distal UC (proctitis, proctosigmoiditis)	2 mg BID × 2 weeks, then daily × 4 weeks
Hydrocortisone enema / foam	Distal UC	100 mg enema at bedtime
Prednisone (oral)	**Moderate–severe** UC or CD flare (bridge to maintenance therapy)	**40–60 mg daily** until response (1–2 weeks), then **taper over 8–12 weeks** (e.g., ↓ 5 mg/week)

Drug	Use	Dose
Methylprednisolone IV	**Acute severe UC** / severe CD (hospitalized)	**60 mg/day** (or **hydrocortisone 100 mg IV q8h**)

- **Budesonide: high first-pass metabolism** (~90%, **CYP3A4**) → **fewer systemic ADRs • Avoid strong CYP3A4 inhibitors** (ketoconazole, ritonavir, clarithromycin, **grapefruit juice**) → ↑ systemic steroid effects (Cushing's).
- **Systemic steroid ADRs: hyperglycemia**, **hypertension**, **osteoporosis** (give **calcium + vitamin D**; bone density), **infections**, **adrenal suppression** (**taper**, don't stop abruptly), mood changes, insomnia, weight gain, **cataracts, glaucoma**, acne, poor wound healing, avascular necrosis.
- **Steroid-dependent** (flare when tapering) or **steroid-refractory** → start an **immunomodulator or biologic**.
- **Never use steroids for maintenance.**

7. Immunomodulators

Thiopurines: azathioprine and 6-mercaptopurine (6-MP)

Point	Details
Use	**Maintenance** of remission (UC and CD) • **Steroid-sparing** • Combined with **infliximab** (↓ anti-drug antibodies; **SONIC trial**) • **Not for induction alone** (slow onset **3–6 months**)
Dose	**Azathioprine 2–2.5 mg/kg/day** • **6-MP 1–1.5 mg/kg/day** (azathioprine is a prodrug of 6-MP)
Before starting	Check **TPMT** (and **NUDT15**) genotype/activity • **Low/intermediate** → reduce dose • **Absent** → **avoid** (severe myelosuppression)
ADRs	**Myelosuppression** (leukopenia) • **Hepatotoxicity** • **Pancreatitis** (early, hypersensitivity → stop) • Nausea • **Infections** • **Lymphoma** • **Non-melanoma skin cancer** (sun protection, yearly skin exams) • **Hepatosplenic T-cell lymphoma** (rare, fatal; **young men** on **thiopurine + anti-TNF**)
Monitoring	**CBC and LFTs** frequently at start (every 1–2 weeks), then every 3 months • 6-TGN/6-MMP metabolite levels
DDIs	**Allopurinol and febuxostat** (xanthine oxidase inhibitors) → ↑ 6-MP levels → **severe myelosuppression** → **reduce thiopurine dose to 25–33%** (avoid febuxostat) • **5-ASA** → ↑ myelosuppression • ↓ **warfarin** effect • ACE inhibitors, TMP-SMX → ↑ myelosuppression
Pregnancy	Can be **continued**

Methotrexate

Point	Details
Use	**Crohn's disease** (induction and maintenance; steroid-sparing) • With anti-TNF to ↓ immunogenicity • **Not effective for UC**
Dose	Induction **25 mg IM/SC once weekly** • Maintenance **15 mg once weekly** • **Once WEEKLY** (daily dosing errors are fatal)
Folic acid	**1 mg daily** (or 5 mg weekly) to ↓ ADRs
ADRs	**Hepatotoxicity** (fibrosis; avoid alcohol) • **Myelosuppression** • **Pneumonitis** • Mouth ulcers, nausea • **Teratogenic** → **contraindicated in pregnancy**; stop **≥ 3 months before conception** (men and women)
Monitoring	CBC, LFTs, SCr every 1–3 months
DDIs	**TMP-SMX** (↑ myelosuppression — avoid) • **NSAIDs**, **PPIs**, penicillins, probenecid (↓ clearance) • Alcohol

Calcineurin inhibitors (rescue therapy)

- **Cyclosporine IV 2–4 mg/kg/day** (continuous infusion) for **acute severe UC** not responding to IV steroids — **bridge** to thiopurine or vedolizumab maintenance. Tacrolimus is an alternative.

- **ADRs: nephrotoxicity**, **hypertension**, **hypomagnesemia**, **seizures** (↑ risk with low Mg/cholesterol), tremor, hyperkalemia, gingival hyperplasia, hirsutism, infections (**PJP prophylaxis** with TMP-SMX).
- **Monitor trough levels • CYP3A4/P-gp** interactions (azoles, macrolides, grapefruit ↑; rifampin, phenytoin, St. John's wort ↓).

8. Biologics

Anti-TNF agents

Drug	Indication	Dose
Infliximab (Remicade; biosimilars) — **IV**, chimeric	**UC and CD** (including **fistulizing CD**), acute severe UC rescue	**5 mg/kg IV at weeks 0, 2, 6**, then **every 8 weeks** (up to 10 mg/kg or every 4 weeks if loss of response) • SC maintenance form available
Adalimumab (Humira; biosimilars) — SC	**UC and CD**	**160 mg week 0**, **80 mg week 2**, then **40 mg every 2 weeks** (may ↑ to weekly)
Certolizumab pegol (Cimzia) — SC	**CD only**	**400 mg at weeks 0, 2, 4**, then **every 4 weeks • PEGylated, no Fc → minimal placental transfer** (good in pregnancy)
Golimumab (Simponi) — SC	**UC only**	**200 mg week 0**, **100 mg week 2**, then **100 mg every 4 weeks**

Point	Details
Boxed warnings	**Serious infections** (including **TB reactivation**, fungal, opportunistic) • **Malignancy** (**lymphoma**, **hepatosplenic T-cell lymphoma** with thiopurines)
Screen before starting	**TB** (**IGRA or PPD** + chest X-ray if positive → treat latent TB first) • **Hepatitis B** (HBsAg, anti-HBc, anti-HBs) • Hepatitis C, HIV
Contraindications	**Active infection • Moderate–severe heart failure (NYHA III–IV)** — worsens HF • **Demyelinating disease** (MS, optic neuritis)
Other ADRs	**Infusion reactions** (infliximab; premedicate with acetaminophen, antihistamine ± steroid) • **Injection site reactions • Drug-induced lupus • Paradoxical psoriasis** • Hepatotoxicity • Cytopenias • **Melanoma**
Immunogenicity	Anti-drug antibodies → **loss of response** and infusion reactions • ↓ by **combining with thiopurine or methotrexate** • Use **therapeutic drug monitoring** (trough levels and antibodies)
Vaccines	**No live vaccines** during therapy • Infants exposed in utero (especially 3rd trimester): **avoid live vaccines in the first 6 months** (e.g., BCG; rotavirus per current guidance)

Anti-integrin agents

Drug	Key points
Vedolizumab (Entyvio)	**Gut-selective α4β7** integrin blocker → blocks lymphocyte trafficking to the **gut only** • **UC and CD • 300 mg IV at weeks 0, 2, 6**, then **every 8 weeks** (SC maintenance 108 mg every 2 weeks available) • **Best safety profile** (minimal systemic immunosuppression) — good for **elderly** and infection/cancer risk • **Slower onset** • No PML signal • ADRs: nasopharyngitis, headache, arthralgia, infusion reactions
Natalizumab (Tysabri)	α4 integrin (not gut-selective) • **CD only** (rarely used) • **Boxed warning: PML** (JC virus) → restricted program (TOUCH), test **JCV antibodies** • **Do not combine** with immunosuppressants or anti-TNF

Interleukin inhibitors

Drug	Target	Indication / dose
Ustekinumab (Stelara; biosimilars)	**IL-12 and IL-23** (p40)	**UC and CD** • **Single weight-based IV induction (~6 mg/kg)**: 260 mg (≤ 55 kg), 390 mg (55–85 kg), 520 mg (> 85 kg) • Then **90 mg SC every 8 weeks** • Infections, nasopharyngitis; good safety
Risankizumab (Skyrizi)	**IL-23** (p19)	**UC and CD** • IV induction at weeks 0, 4, 8 (CD 600 mg; UC 1,200 mg) → SC maintenance every 8 weeks (on-body injector) • **Monitor LFTs** (hepatotoxicity)
Mirikizumab (Omvoh)	**IL-23** (p19)	**UC and CD** • IV induction (UC 300 mg every 4 weeks × 3) → SC maintenance every 4 weeks • LFTs
Guselkumab (Tremfya)	**IL-23** (p19)	**UC and CD** • IV (or SC) induction → SC maintenance every 4–8 weeks

- All biologics: **screen for TB and hepatitis B**, avoid **live vaccines**, watch for **infections**.

9. Oral Small Molecules

JAK inhibitors

Drug	Indication	Dose
Tofacitinib (Xeljanz)	**UC only**	**10 mg BID × 8 weeks** (up to 16) → **5 mg BID** maintenance (10 mg BID if needed)
Upadacitinib (Rinvoq)	**UC and CD**	UC: **45 mg daily × 8 weeks** → 15–30 mg daily • CD: **45 mg daily × 12 weeks** → 15–30 mg daily

- **Fast onset** (days) • Oral • Not immunogenic.
- **Boxed warnings: serious infections**, **mortality**, **malignancy** (lymphoma, lung cancer), **MACE** (MI, stroke), **thrombosis** (**VTE, PE**, arterial).
- **FDA:** use only after **failure or intolerance of ≥ 1 TNF blocker**.
- **Other ADRs: herpes zoster** (give **recombinant zoster vaccine, Shingrix**, before starting) • ↑ **LDL/lipids** • cytopenias (lymphopenia, neutropenia, anemia) • ↑ LFTs, ↑ CK • acne (upadacitinib).
- **DDIs: strong CYP3A4 inhibitors** (ketoconazole) → **reduce dose** (tofacitinib 10 mg BID → 5 mg BID; 5 mg BID → 5 mg daily) • **Strong CYP3A4 inducers** (rifampin) → **avoid** • **Do not combine** with biologics or potent immunosuppressants (azathioprine, cyclosporine).
- **Avoid in pregnancy** (contraception) • Monitor CBC, LFTs, lipids.

S1P receptor modulators (UC only)

Drug	Dose
Ozanimod (Zeposia)	**Titration:** 0.23 mg daily days 1–4 → 0.46 mg days 5–7 → **0.92 mg daily**
Etrasimod (Velsipity)	**2 mg once daily** (no titration)

- **Mechanism:** trap lymphocytes in lymph nodes → fewer reach the gut.
- **ADRs: bradycardia and AV block** at the **first dose** (titration; **baseline ECG**) • **Macular edema** (eye exam if diabetes/uveitis) • ↑ **LFTs** • **Infections** (lymphopenia; check **VZV immunity**) • ↑ blood pressure • PML, PRES (rare).
- **Contraindications:** recent **MI, stroke, unstable angina**, **NYHA III–IV heart failure** (within 6 months) • **2nd/3rd-degree AV block** or sick sinus without pacemaker • Severe untreated sleep apnea • **Ozanimod + MAO inhibitors** (contraindicated).
- **DDIs:** ozanimod with **MAOIs**, adrenergic/serotonergic drugs (caution: hypertensive crisis risk), **tyramine-rich foods** (large amounts) • Beta-blockers/CCBs (↑ bradycardia) • Strong CYP2C8 inhibitors (gemfibrozil) ↑ ozanimod.
- **Pregnancy:** use contraception during and for 3 months after.

10. Antibiotics and Symptomatic Therapy

Agent	Use / key points
Metronidazole	**Perianal CD** (fistulas, abscesses), **pouchitis** • 10–20 mg/kg/day • Peripheral neuropathy (long term), metallic taste, **no alcohol**
Ciprofloxacin	**Perianal CD**, **pouchitis** • 500 mg BID • Tendinopathy, QT
Loperamide, diphenoxylate/atropine	Mild symptomatic diarrhea in **mild/inactive** disease • **Avoid in active severe colitis** (risk of **toxic megacolon**)
Cholestyramine	**Bile-acid diarrhea** after **ileal resection** (CD) • Binds other drugs → separate (1 h before / 4–6 h after)
Acetaminophen	**Preferred analgesic** • **Avoid NSAIDs** (may trigger flares)
Avoid	**Opioids** and **anticholinergics** in active disease (toxic megacolon, masking symptoms) • **NSAIDs**
Probiotics	**Not recommended** for UC or CD treatment (may help **pouchitis**)

11. Treatment Algorithm: Ulcerative Colitis

Situation	Induction	Maintenance
Mild–moderate proctitis	**Mesalamine suppository 1 g daily** (first-line) • Rectal steroids if 5-ASA not tolerated	**Mesalamine suppository 1 g daily** (or 3 times weekly)
Mild–moderate left-sided colitis	**Mesalamine enema** (1–4 g) + **oral mesalamine ≥ 2 g/day** (combination best) • **Budesonide MMX 9 mg** if 5-ASA fails	Oral mesalamine ± rectal mesalamine
Mild–moderate extensive colitis	**Oral mesalamine 2–4.8 g/day** (± rectal) • Budesonide MMX or prednisone if no response	**Oral mesalamine ≥ 2 g/day**
Moderate–severe UC	**Oral prednisone** (short bridge) **or** start an advanced therapy: **anti-TNF** (infliximab ± thiopurine, adalimumab, golimumab), **vedolizumab**, **ustekinumab**, **IL-23 inhibitors**, **JAK inhibitors** (after TNF), **S1P modulators**	**Continue the agent that induced remission** • After steroid induction → thiopurine or advanced therapy • **Not steroids**
Acute severe UC (hospitalized)	**IV methylprednisolone 60 mg/day** (or hydrocortisone 100 mg q8h) • **VTE prophylaxis** (heparin) • **Test for C. difficile** • Avoid opioids, anticholinergics, antidiarrheals, NSAIDs • **No response in 3–5 days** → **infliximab** or **cyclosporine** (rescue) → **colectomy** if fails	Infliximab, or thiopurine/vedolizumab after cyclosporine
Toxic megacolon	Emergency: bowel rest, IV steroids, IV antibiotics, surgery consult	—

Colectomy (with ileal pouch-anal anastomosis, IPAA) **cures UC**. A common complication is **pouchitis** → **ciprofloxacin** or **metronidazole**.

12. Treatment Algorithm: Crohn's Disease

Situation	Treatment
Mild–moderate ileocecal CD	**Budesonide (Entocort EC) 9 mg daily × 8 weeks** (induction) • **5-ASA not recommended** (sulfasalazine may help mild colonic CD only)
Moderate–severe CD	**Oral prednisone** for short-term induction (bridge) • **Early advanced therapy** (especially high-risk): **anti-TNF** (infliximab, adalimumab, certolizumab) ± **thiopurine** (combination is most effective), **ustekinumab**, **vedolizumab**, **risankizumab/IL-23 inhibitors**, **upadacitinib** (after TNF failure)
Maintenance	**Continue the inducing biologic** • **Thiopurines** or **methotrexate** (steroid-sparing) • **Not steroids, not 5-ASA**

Situation	Treatment
Severe / fulminant CD	**IV corticosteroids** • Rule out abscess (drain before immunosuppression) • Anti-TNF • Surgery
Fistulizing / perianal CD	**Antibiotics** (**metronidazole**, **ciprofloxacin**) + **drainage** (seton) • **Infliximab** (most evidence for fistula closure) ± thiopurine; adalimumab
Stricturing CD	Endoscopic dilation or surgery (strictureplasty, resection)
After surgical resection	Prevent recurrence: **anti-TNF** (± thiopurine) in high-risk patients • **Stop smoking** • Colonoscopy at 6–12 months

Smoking cessation is one of the most important interventions in **Crohn's disease**. Surgery does **not** cure CD.

13. Health Maintenance and Pre-Treatment Screening

Area	Recommendation
Before biologics / small molecules	**TB** (IGRA or PPD) • **Hepatitis B** (HBsAg, anti-HBc, anti-HBs) and C • HIV • **VZV immunity** • **TPMT** before thiopurines • Baseline CBC, LFTs, SCr • ECG before S1P modulators • Lipids before JAK inhibitors
Inactivated vaccines (safe on therapy)	**Influenza** (inactivated) yearly • **Pneumococcal** • **Hepatitis B** • **HPV** • **Recombinant zoster (Shingrix)** • COVID-19, Tdap
Live vaccines	**Avoid** during immunosuppression (MMR, varicella, live zoster, intranasal flu, yellow fever) • Give **≥ 4 weeks before** starting therapy • Immunosuppression includes **prednisone ≥ 20 mg/day for ≥ 2 weeks**, thiopurines, MTX, biologics, JAK inhibitors
Colorectal cancer surveillance	**Colonoscopy 8 years after diagnosis** (UC beyond rectum, or CD involving ≥ 1/3 of colon), then every 1–3 years • **Yearly** if **PSC** (start at PSC diagnosis)
Skin cancer	Yearly skin exam (thiopurines → non-melanoma; anti-TNF → melanoma) • Sun protection
Cervical cancer	Regular screening in women on immunosuppressants
Bone health	**Calcium + vitamin D** with steroids • Bone density (DEXA) if steroid use, low BMI, postmenopausal
Nutrition	**Iron** (IV iron preferred if active disease or oral intolerance), **B12** (ileal CD/resection), **vitamin D**, folate (sulfasalazine, MTX), zinc
VTE	**Pharmacologic VTE prophylaxis** for all hospitalized IBD patients (even with rectal bleeding)
Mental health	Screen for **depression and anxiety**

14. Special Populations

Population	Key points
Pregnancy	**Active disease** is the biggest risk → **conceive in remission** and **continue most therapies** • **Safe/continue:** 5-ASA (sulfasalazine + **folic acid 2 mg**), **thiopurines**, **anti-TNF** (certolizumab has minimal placental transfer), **vedolizumab**, **ustekinumab**, steroids if needed • **Contraindicated: methotrexate** (stop ≥ 3 months before), **thalidomide** • **Avoid: JAK inhibitors**, **S1P modulators** • Avoid Asacol products with **dibutyl phthalate** coating
Breastfeeding	Most therapies compatible (5-ASA, thiopurines, anti-TNF, vedolizumab, ustekinumab) • Avoid **MTX**, JAK inhibitors, S1P modulators
Elderly	Higher infection and cancer risk → **vedolizumab** and **ustekinumab** attractive • Minimize steroids • Avoid anti-TNF in HF
Children	Growth failure (CD) • **Exclusive enteral nutrition** for induction in pediatric CD • Avoid long-term steroids

Exam Traps and Frequently Tested Points

1. **UC:** colon only, starts at **rectum**, **continuous**, **mucosal**, **bloody diarrhea**, p-ANCA, **colectomy curative**.
2. **CD:** mouth to anus (**terminal ileum**), **skip lesions**, **transmural**, **fistulas/strictures**, granulomas, ASCA.
3. **Smoking protects in UC** but **worsens CD → stop smoking** in CD.
4. **5-ASA** = first-line for **mild–moderate UC** (induction + maintenance); **not for CD**.
5. **Proctitis → mesalamine suppository • Left-sided → enema + oral • Extensive → oral** (± rectal).
6. **Rectal 5-ASA > rectal steroids**; **oral + rectal > oral alone**.
7. **Sulfasalazine:** sulfa allergy, **G6PD hemolysis**, **agranulocytosis**, **oligospermia**, **orange urine/skin**, **folate deficiency → folic acid**.
8. **Mesalamine: interstitial nephritis** → monitor SCr; paradoxical colitis.
9. **Steroids = induction only, never maintenance** • taper; calcium + vitamin D.
10. **Budesonide:** Entocort → **ileocecal CD**; Uceris MMX → **UC**; high first-pass; **avoid CYP3A4 inhibitors/grapefruit**.
11. **Acute severe UC: IV methylprednisolone 60 mg/day** → no response in **3–5 days** → **infliximab or cyclosporine** → colectomy.
12. **Hospitalized IBD → VTE prophylaxis** (even with bleeding) • test for **C. difficile** in flares.
13. **Avoid antidiarrheals, opioids, anticholinergics** in severe colitis (**toxic megacolon**) • **avoid NSAIDs** (use acetaminophen).
14. **Thiopurines:** slow (3–6 months) → **maintenance • check TPMT/NUDT15** • myelosuppression, pancreatitis, lymphoma, skin cancer.
15. **Allopurinol/febuxostat + azathioprine/6-MP** → severe myelosuppression → **reduce to 25–33%**.
16. **Methotrexate: CD only**, **once weekly**, folic acid, **teratogenic** (stop 3 months before conception), avoid TMP-SMX.
17. **Cyclosporine:** rescue in acute severe UC • nephrotoxicity, HTN, hypomagnesemia, seizures • monitor levels.
18. **Anti-TNF: screen TB and HBV • contraindicated in NYHA III–IV HF** and **demyelinating disease** • drug-induced lupus, psoriasis.
19. **Certolizumab = CD only**, minimal placental transfer • **Golimumab = UC only** • **Infliximab** = best for **fistulas**.
20. **Infliximab + thiopurine** (combination) → more effective, ↓ antibodies; watch **HSTCL** in young men.
21. **Vedolizumab: gut-selective**, safest, slower onset • **Natalizumab:** PML (JC virus).
22. **Ustekinumab:** IL-12/23, **single weight-based IV dose** then SC every 8 weeks • **Risankizumab:** monitor LFTs.
23. **JAK inhibitors** (tofacitinib = UC only; upadacitinib = UC + CD): **boxed: infection, death, cancer, MACE, thrombosis • after ≥ 1 TNF failure • herpes zoster** (Shingrix) • lipids.
24. **Tofacitinib + strong CYP3A4 inhibitor** → reduce dose; avoid with strong inducers.
25. **S1P modulators** (ozanimod, etrasimod; UC only): **first-dose bradycardia**, **macular edema**, LFTs • ozanimod + **MAOIs** contraindicated.
26. **Perianal CD** → metronidazole/ciprofloxacin + drainage + infliximab • **Pouchitis** → cipro/metronidazole.
27. **No live vaccines** on immunosuppressants (prednisone ≥ 20 mg ≥ 2 weeks) • give inactivated vaccines + Shingrix.
28. **Colonoscopy surveillance 8 years** after diagnosis; yearly with **PSC**.
29. **Pregnancy:** continue 5-ASA, thiopurines, anti-TNF, vedolizumab, ustekinumab • **MTX contraindicated** • avoid JAKi and S1P.
30. **Ileal CD/resection → B12** deficiency, **bile-acid diarrhea** (cholestyramine), **oxalate kidney stones**, gallstones.

> **Note:** Doses and preferred agents differ between references and are updated often (new biologics are added frequently). **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Mesalamine (oral)	Induction 2.4–4.8 g/day • Maintenance ≥ 2 g/day (Lialda once daily)	Headache, nausea, **interstitial nephritis**, pancreatitis, paradoxical colitis	Monitor SCr • ↑ myelosuppression with thiopurines • pH-dependent forms: antacids/PPIs may cause early release • **UC only**
Mesalamine (rectal)	Suppository 1 g HS (proctitis) • Enema 4 g HS (left-sided)	Local irritation, anal discomfort	Retain enema ≥ 8 h • suppository stains fabrics
Sulfasalazine	Induction 4–6 g/day • Maintenance 2–4 g/day (divided, with food)	Nausea, headache, **rash (sulfa)**, **G6PD hemolysis**, **agranulocytosis**, hepatotoxicity, **oligospermia**, **orange urine/skin**	↓ **folate** (give folic acid) • ↓ digoxin absorption • ↑ thiopurine toxicity • stains contact lenses
Balsalazide	2.25 g TID	Headache, abdominal pain	Colon-targeted prodrug
Olsalazine	500 mg BID	**Secretory diarrhea**	Colon-targeted
Budesonide EC (Entocort)	9 mg daily × 8 wk (ileocecal CD)	Fewer systemic steroid effects; adrenal suppression with long use	**CYP3A4 substrate** → avoid ketoconazole, ritonavir, **grapefruit**
Budesonide MMX (Uceris)	9 mg daily × 8 wk (UC)	Same	Same CYP3A4 DDIs
Prednisone	40–60 mg daily, taper over 8–12 wk	**Hyperglycemia**, HTN, **osteoporosis**, infection, **adrenal suppression**, mood, cataracts	Induction only • calcium + vitamin D • taper • ↑ GI bleed with NSAIDs
Methylprednisolone IV	60 mg/day (acute severe UC)	Same as systemic steroids	Reassess at 3–5 days
Hydrocortisone	100 mg IV q8h • enema 100 mg HS	Same	Rectal for distal UC
Azathioprine	2–2.5 mg/kg/day	**Myelosuppression**, **pancreatitis**, hepatotoxicity, **lymphoma**, **NMSC**, HSTCL	**Check TPMT/NUDT15** • **allopurinol/febuxostat** → ↓ dose to 25–33% • 5-ASA, TMP-SMX, ACEi ↑ myelosuppression • ↓ warfarin
6-Mercaptopurine	1–1.5 mg/kg/day	Same as azathioprine	Same as azathioprine
Methotrexate	25 mg IM/SC weekly (induction) → 15 mg weekly (maintenance)	**Hepatotoxicity**, myelosuppression, **pneumonitis**, mucositis, **teratogenic**	**CD only** • **once weekly** • folic acid • avoid **TMP-SMX**, NSAIDs, PPIs, alcohol • stop 3 months before conception
Cyclosporine	2–4 mg/kg/day IV (acute severe UC rescue)	**Nephrotoxicity**, **HTN**, **hypomagnesemia**, **seizures**, tremor, gingival hyperplasia, infections	**CYP3A4/P-gp** (azoles, macrolides, grapefruit ↑; rifampin, St. John's wort ↓) • monitor levels • PJP prophylaxis
Infliximab	5 mg/kg IV wk 0, 2, 6 → q8w	**Boxed: serious infections (TB), malignancy • infusion reactions**, lupus-like, psoriasis	**Screen TB, HBV • CI: NYHA III–IV HF, demyelinating disease** • no live vaccines • combine with thiopurine
Adalimumab	160 mg wk 0 → 80 mg wk 2 → 40 mg q2w SC	Same boxed warnings • injection site reactions	Same as infliximab
Certolizumab pegol	400 mg SC wk 0, 2, 4 → q4w	Same boxed warnings	**CD only** • minimal placental transfer
Golimumab	200 mg SC wk 0 → 100 mg wk 2 → 100 mg q4w	Same boxed warnings	**UC only**

Drug	Dose	Key ADRs	Key DDIs / special notes
Vedolizumab	300 mg IV wk 0, 2, 6 → q8w (or 108 mg SC q2w)	Nasopharyngitis, headache, arthralgia, infusion reactions	**Gut-selective** • safest infection profile • slower onset • UC and CD
Natalizumab	300 mg IV q4w	**Boxed: PML**	CD only • JCV antibody testing • no combination with immunosuppressants
Ustekinumab	~6 mg/kg IV once (260/390/520 mg) → 90 mg SC q8w	Infections, nasopharyngitis	Screen TB, HBV • UC and CD
Risankizumab	IV wk 0, 4, 8 (CD 600 mg / UC 1,200 mg) → SC q8w	Infections, **hepatotoxicity**	Monitor **LFTs** • UC and CD
Mirikizumab	UC: 300 mg IV q4w × 3 → 200 mg SC q4w	Infections, ↑ LFTs, injection site reactions	UC and CD
Guselkumab	IV/SC induction → SC q4–8w	Infections, ↑ LFTs	UC and CD
Tofacitinib	10 mg BID × 8 wk → 5 mg BID	**Boxed: infection, death, malignancy, MACE, thrombosis • herpes zoster**, ↑ lipids	**UC only** • after TNF failure • **strong CYP3A4 inhibitors → ↓ dose**; avoid strong inducers • no biologics/azathioprine
Upadacitinib	UC 45 mg × 8 wk / CD 45 mg × 12 wk → 15–30 mg daily	Same boxed warnings • zoster, acne, ↑ lipids, ↑ CK	UC and CD • CYP3A4 DDIs • avoid in pregnancy
Ozanimod	0.23 mg d1–4 → 0.46 mg d5–7 → 0.92 mg daily	**Bradycardia/AV block** (first dose), **macular edema**, ↑ LFTs, infections, ↑ BP	**UC only** • **CI with MAOIs** • caution beta-blockers/CCBs • baseline ECG • gemfibrozil ↑
Etrasimod	2 mg daily	Same as ozanimod	**UC only** • baseline ECG • CYP2C8/2C9/3A4 inhibitors ↑
Metronidazole	10–20 mg/kg/day (perianal CD) • 500 mg BID–TID	Metallic taste, **neuropathy**	**No alcohol** • ↑ warfarin INR
Ciprofloxacin	500 mg BID	Tendinopathy, QT, CNS	Chelation (separate from antacids, iron, Ca) • ↑ theophylline, tizanidine (CI)
Loperamide	4 mg then 2 mg after each loose stool (max 16 mg/day)	Constipation, abdominal cramps	**Avoid in severe colitis** (toxic megacolon) • QT at high doses
Cholestyramine	4 g 1–4 times daily	Constipation, bloating	Binds drugs → take others 1 h before / 4–6 h after • ↓ fat-soluble vitamins
Folic acid	1 mg daily (sulfasalazine, MTX); 2 mg in pregnancy with sulfasalazine	Rare	Prevents folate deficiency

Good luck on your exam!